

THE PYTHAGOREAN RELATIONSHIP BETWEEN π , e , AND ϕ

Peter Felicetti BE(Civil) BArch(Hons)

¹Director, Felicetti Pty Ltd and Bollinger Grohmann Felicetti Pty Ltd Consulting Engineers,

4th Floor, 145 Russell Street, Melbourne 3000, Australia.

E-mail: peter@felicetti.com.au

Summary: This paper has been prepared to demonstrate a Pythagorean relationship between three transcendental and irrational numbers ϕ , e , and π , creating a triangle with internal angles of 30° , 60° and 90° within an accuracy of 1% in 2 dimensions. and to within 0.02% by incorporating the imaginary number i , $\sqrt{-1}$, in the 3 dimensional relationship.

1. INTRODUCTION

The possible geometric relationship between ϕ , e , and π has always held a fascination with the author as a result of experimentation with deriving novel mathematical proportioning relationships during his engineering and architectural studies. Whilst there are documented cases of relationships between e and π , (Eulers formula $e^{\pi i} + 1 = 0$) and ϕ and π , to date there is no evidence that there has been developed a relationship between all three.

This paper presents the results of the authors investigations and discoveries.

2. π

$$\pi = 3.141592653589793$$

Is an irrational number which means that it cannot be expressed exactly as fraction. It is also a transcendental number. It is the ratio of a circle's circumference to it's diameter.

A transcendental number is described as an irrational number that is not the root of an algebraic equation that has rational coefficients. The number pi cannot be expressed as the root of an equation but only as the limit of some type of infinite process. (1)

An approximation of π to 6 decimal places is 355/113. $\sqrt{10}$ is also an approximation of π

3. ϕ

$$\phi = 1.618033988749894$$

The ubiquity of ϕ in mathematics aroused the interest of many mathematicians in the Middle Ages and during the Renaissance. In 1509 there was published a dissertation by Luca Pacioli, *De Divina Proportione* which was illustrated by Leonardo da Vinci.⁴

The numerical value of phi is calculated by dividing a line AC, such that AB= x and BC=1 so that AB/BC=x=phi

$(x+1)/x = x/1$ i.e. $x^2 - x - 1 = 0$. The positive solution of this is $x = (1 + \sqrt{5})/2 = 1.61803... = \varphi$

Phi is the golden ratio. It is also an irrational and a transcendental number, and reveals itself in the world of nature, associated with phyllotaxis, with the patterns of florets in flowers, with the shape of the nautilus shell and with other natural objects.⁴

4. e

$$e = 2.7182818284590452$$

Is Eulers number. It is an irrational and a transcendental number. The base of natural logarithms (also known as Napierian logarithms, although without historical justification) and the limit of $(1+1/n)^n$ as n approaches infinity. e is an irrational number and is represented by a nondetermining, non repeating decimal. The irrationality of e was proved in 1737 by Euler. Charles Hermite in 1873 proved that e is transcendental: that is, it cannot be a solution of a polynomial equation with integer coefficients. (2)

An approximation of e is $878/323$

5. Imaginary number

The imaginary number is $\sqrt{-1}$. Specifically it is defined as one of the two solutions of the equation

$$X^2 + 1 = 0.$$

Now to solve this equation means to find a number whose square is -1 . Of course, no real number will do, because the square of a real number is never negative. Thus in the domain of real numbers the equation above has no solutions.²

6. Pythagoras

This famous theorem, which we clearly associate with geometry, is also the basis for the field of trigonometry and finds its way into countless other areas, such as art, music, architecture and various fields of mathematics.³

In its most basic sense, the Pythagorean Theorem states that if you draw a square on each side of a right triangle, the sum of the areas of the two smaller squares (i.e. those on the perpendicular sides) is equal to the area of the square drawn on the hypotenuse (the longest side).

This relationship is accredited to the Greek philosopher Pythagoras (ca. 575-495 BCE)

7. Pythagorean triples

Integers that represent the sides of a triangle that have a Pythagorean relationship $a^2 + b^2 = c^2$ are known as Pythagorean triples.³

(3,4,5), (5,12,13), (7, 24, 25) etc.

The smallest triple is (3,4,5). Ancient Egyptian surveyors used ropes with 13 knots evenly spaced, and divided into segments in the ratio 3:4:5 to create a right angle triangle to accurately align their constructions.

8. The Pythagorean relationship between pi, e, and Phi

The author proposes the following relationship;

$$\varphi^2 + e^2 = \pi^2$$

$$\pi^2 = 9.869604397$$

$$\varphi^2 = 2.618033986$$

$$e^2 = 7.389056096$$

$$\varphi^2 + e^2 \cong \pi^2$$

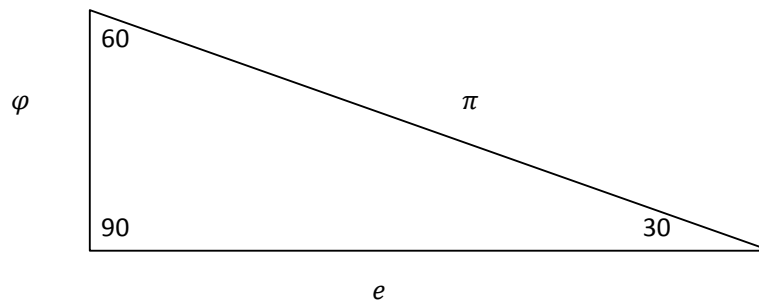
$$2.618033986 + 7.389056096 = 10.00709008$$

$$10.00709008 / 9.869604397 = 1.013930212 \text{ i.e within } 1\%$$

$$\text{Artan}(\varphi/e) = \text{Artan}(0.595241439) = 30.76286101^\circ \text{ i.e approx } 30^\circ$$

$$\text{Artan}(e/\varphi) = \text{Artan}(1.679990561) = 59.23713898^\circ \text{ i.e approx } 60^\circ$$

Ideally,



Further

$$\varphi^2 + e^2 \cong 10$$

$$10.00709008/10 = 1.000709008 \text{ i.e within } 0.1\%$$

$$e - \varphi^2 = 0.100247842$$

$$e - \varphi^2 \cong 1/10$$

$$10(e - \varphi^2) \cong 1$$

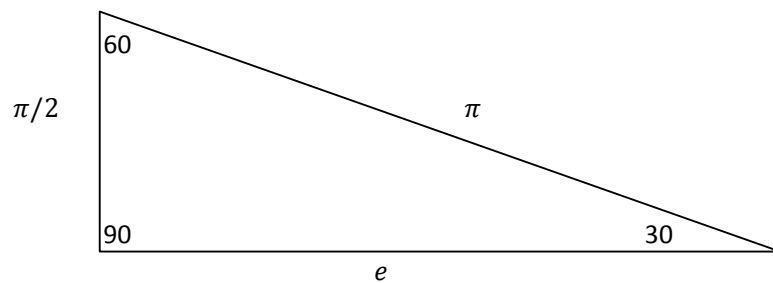
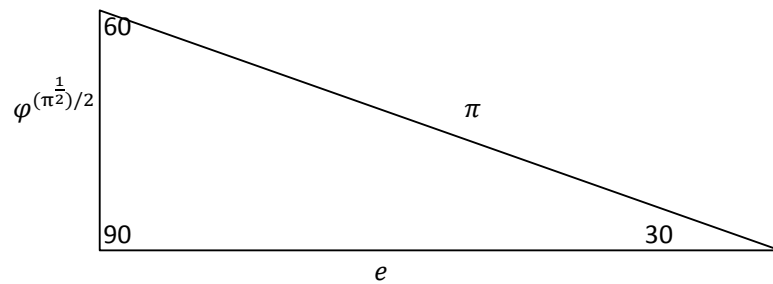
Alternatively, a more precise relationship

$$\varphi^{1.888} + e^2 = \pi^2$$

Approximated to within 1%

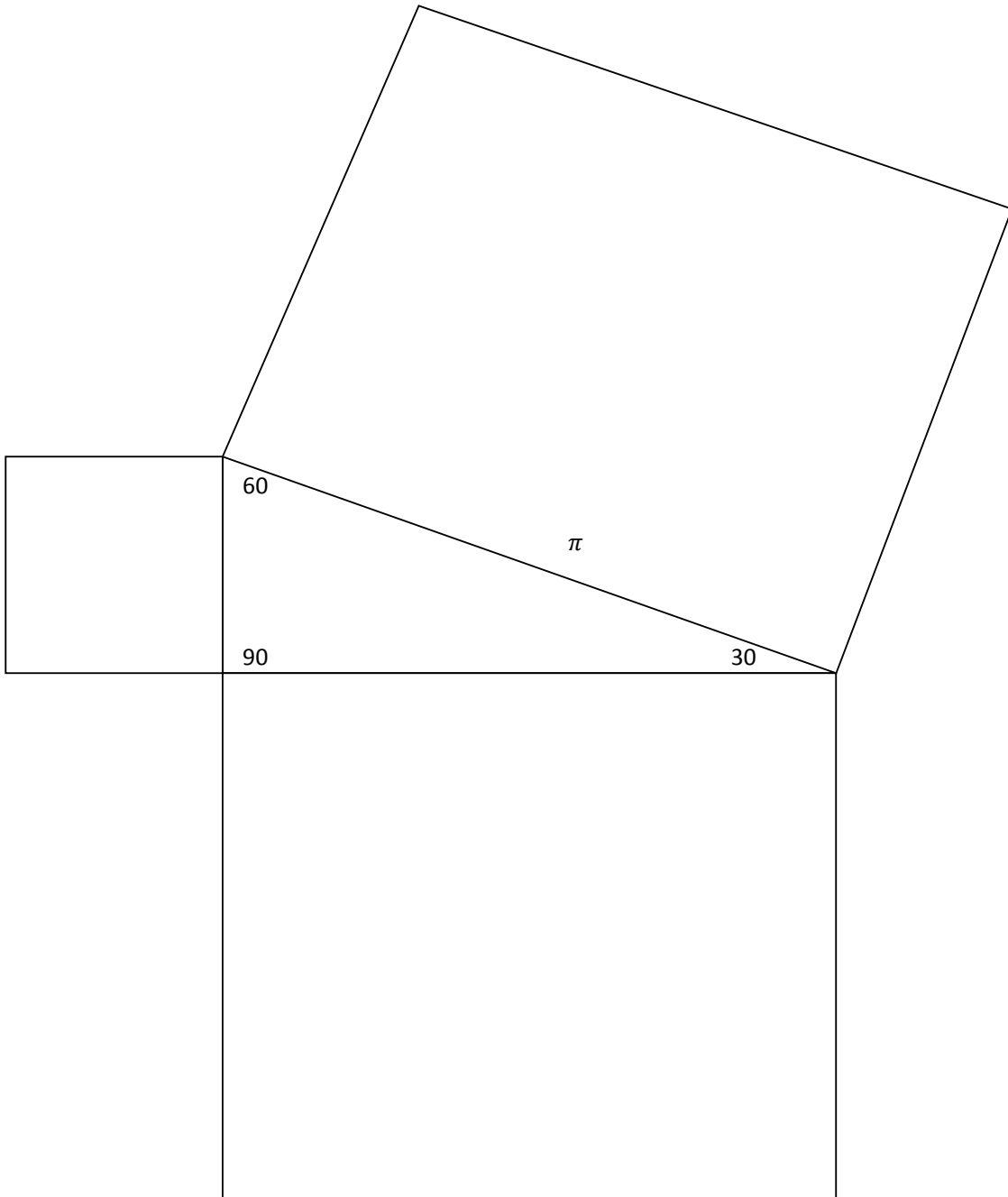
$$\varphi^{\pi^{1/2}} + e^2 = \pi^2$$

9. Alternative approximations



10. Geometrical relationships

In typical Pythagorean fashion when added, the areas of two squares generated by the sides of a right triangle will equal the area of the square of the hypotenuse

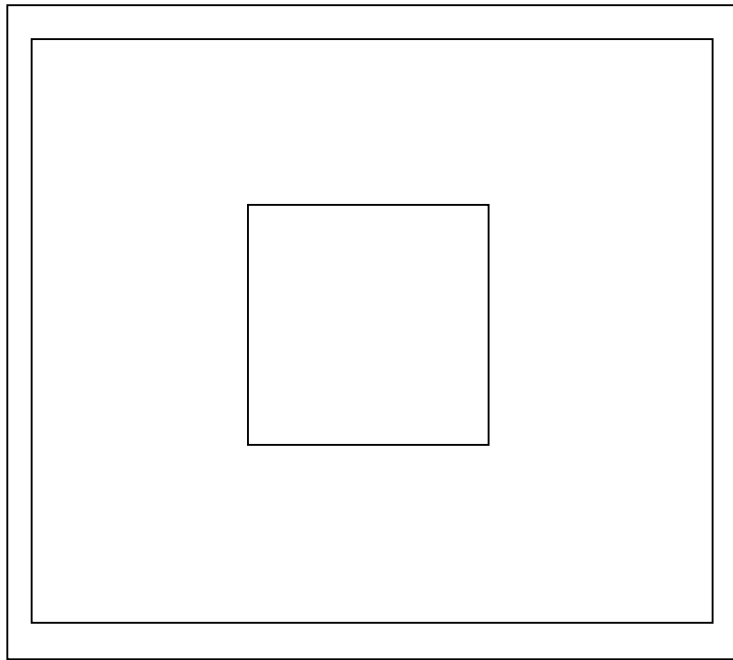


Consider the three squares

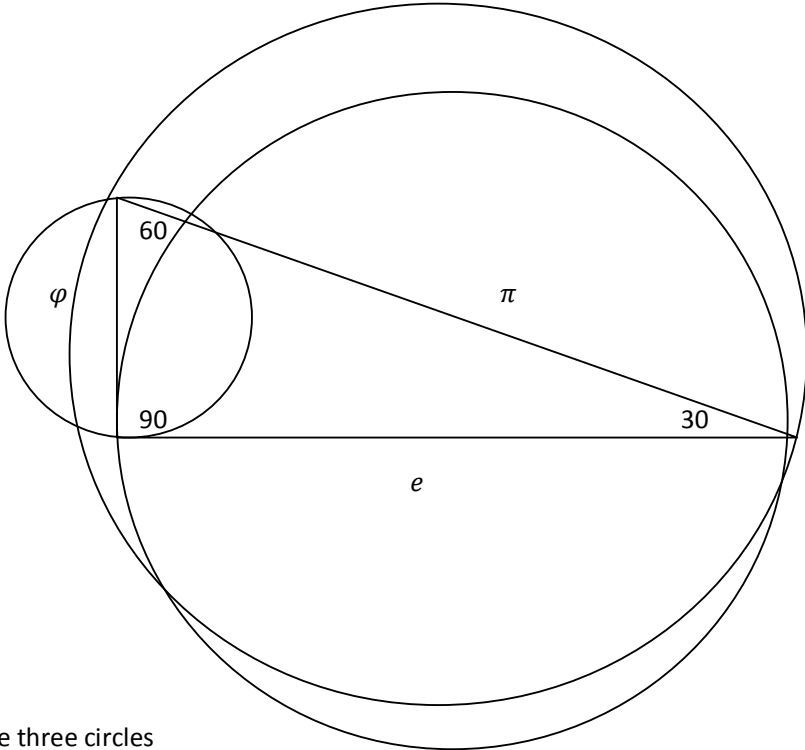
Square A of length φ , Square B of length e , and Square C of length π

The areas of Squares A and B combined will equal the area of Square C in the following relationship

$$\varphi^2 + e^2 = \pi^2$$



Similarly this can be undertaken with any similar geometrical figure generated by the sides of a right triangle₃. For example when added, the areas of two circles generated by the sides of a right triangle will equal the area of the circle of the hypotenuse.

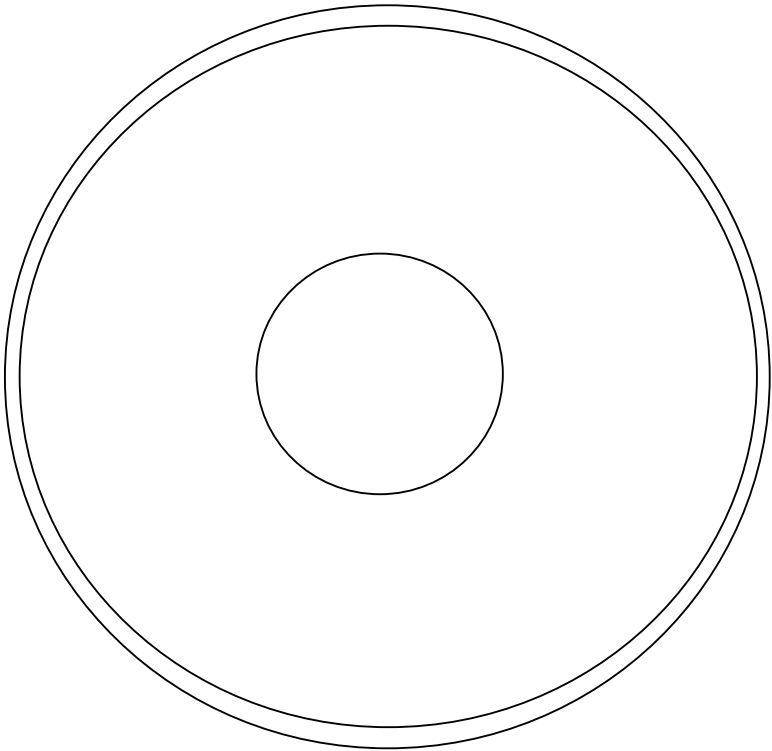


Consider the three circles

Circle A of diameter φ , Circle B of diameter e , and Circle C of diameter π

The areas of Circles A and B combined will equal the area of Circle C in the following relationship

$$(\pi\varphi^2)/4 + (\pi e^2)/4 = (\pi^3)/4$$



11. Pythagorean relationship between φ , e and π , in 3 dimensions

$$a^2 + b^2 + c^2 = d^2$$

$$\varphi^2 + e^2 + z^2 = \pi^2$$

$$2.6179... + 7.389...(\sqrt{-1})^2 \times 0.137$$

Therefore $z = i/2.698$ or i/e where $i = \sqrt{-1}$

To an accuracy of 0.02%

$$\varphi^2 + e^2 + (i/e)^2 = \pi^2$$

CONCLUSION

The author has discovered a pythagorean relationship between transcendental numbers, that will provide some new insights into these numbers.

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